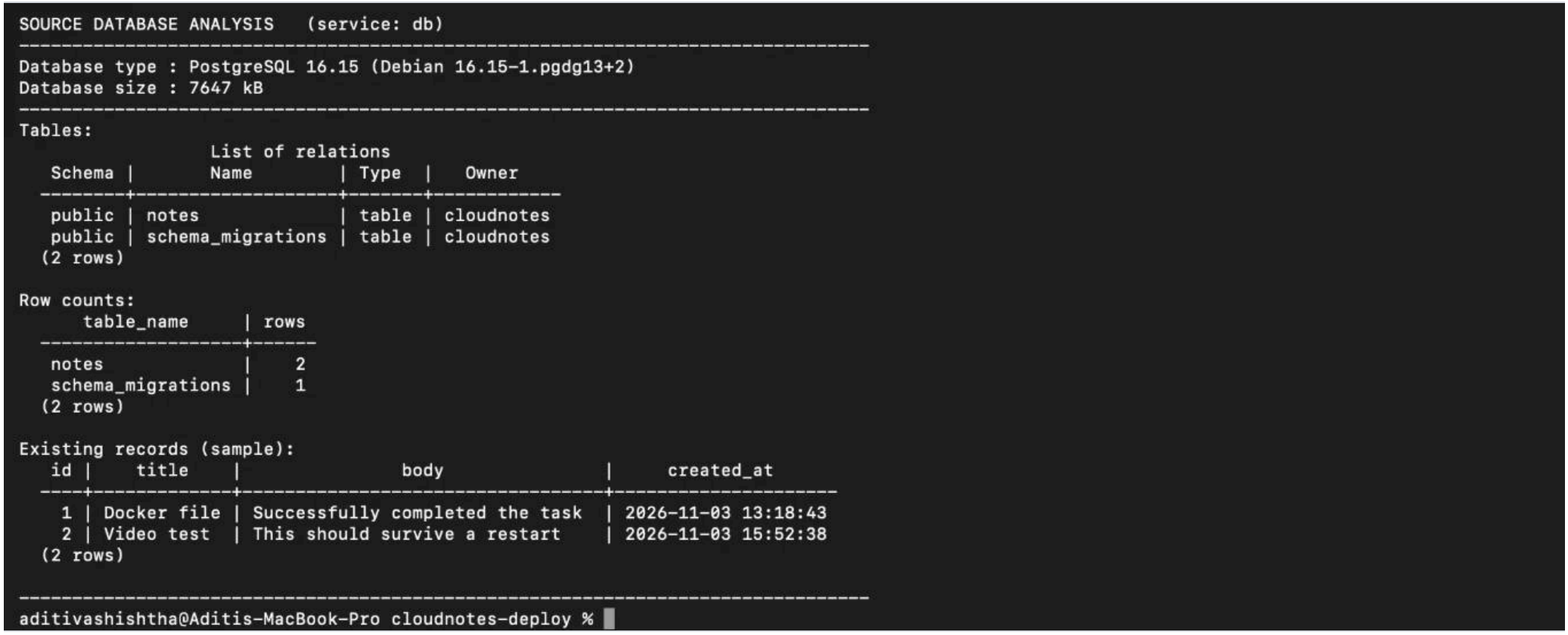
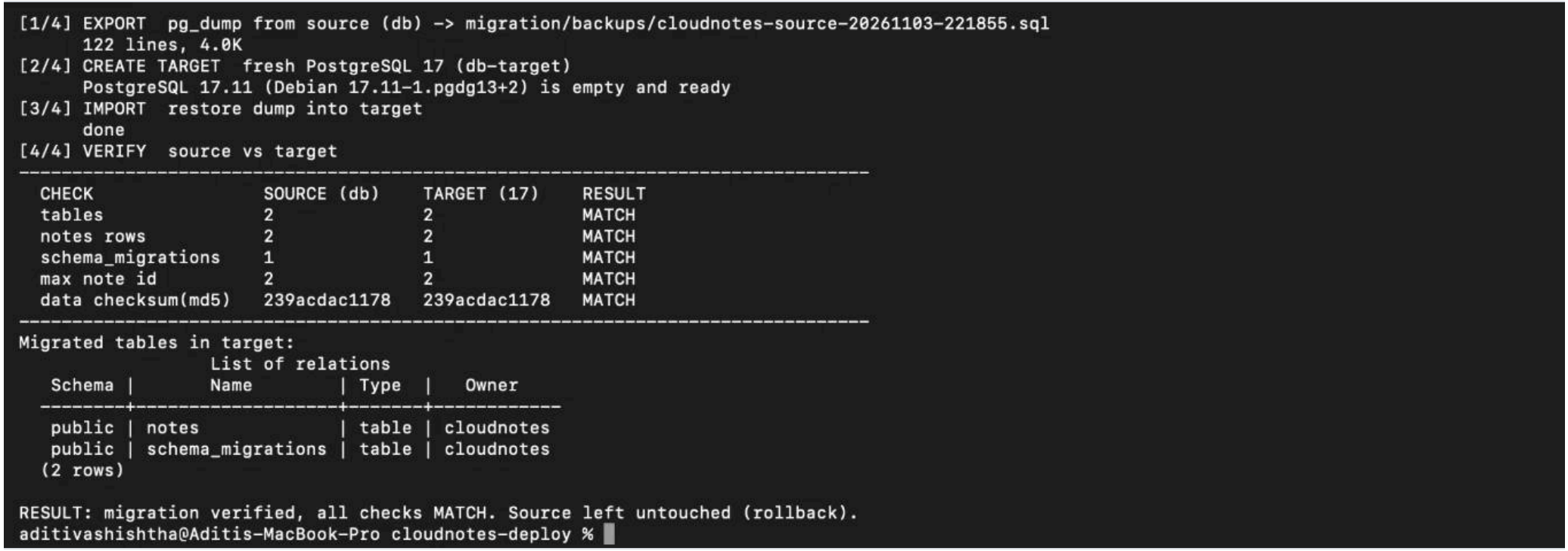
Screenshot 1 — Source Database Analysis



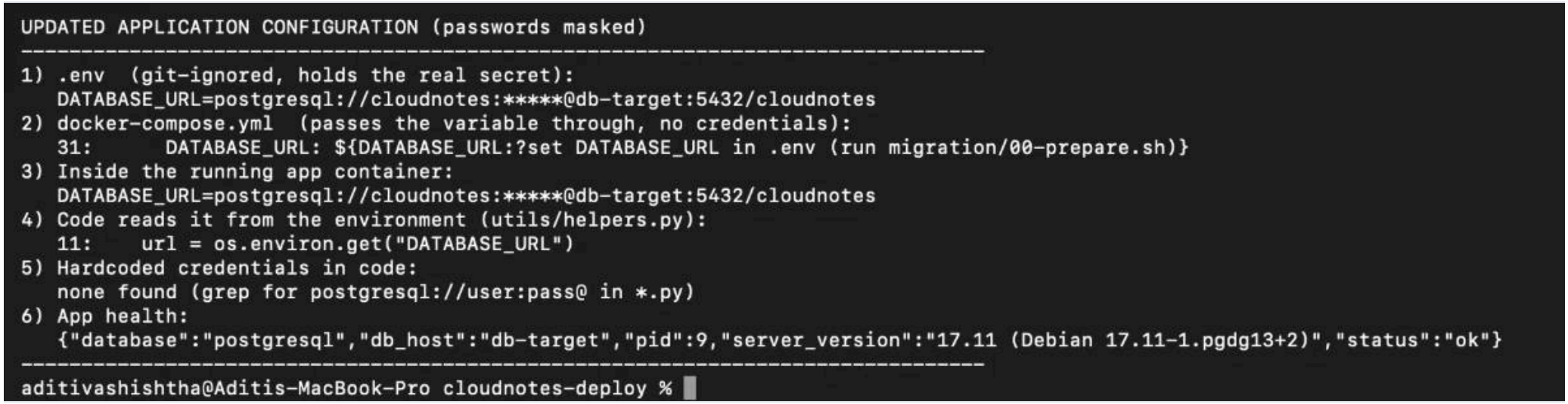
Caption: The source is CloudNotes' existing PostgreSQL 16 database (Docker service 'db'). It has two tables: 'notes' (the user notes: id, title, body, attachment filename, created_at) and 'schema_migrations' (which schema version is applied). The row counts and every existing record are listed above. This is exactly the data that will be migrated, and the source is only read, never changed.

Screenshot 2 — Migrated PostgreSQL Database



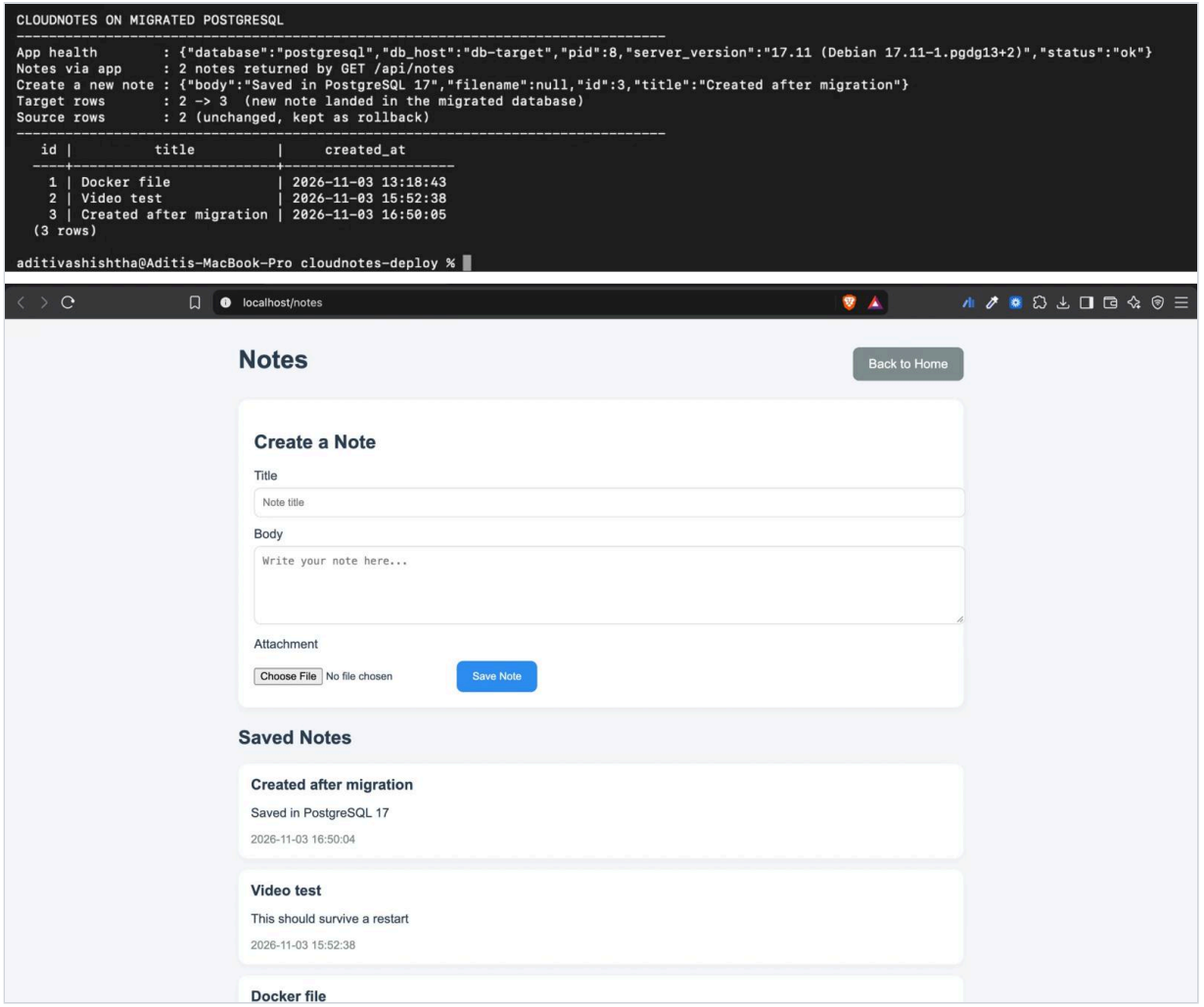
Caption: The data was exported with pg_dump, restored into a brand-new PostgreSQL 17 database (db-target), and then verified by comparing source and target: number of tables, row counts per table, highest note id, and an md5 checksum of every row's contents. All checks MATCH, so the target contains exactly the same records. The source was left untouched as the rollback.

Screenshot 3 — Updated Application Configuration



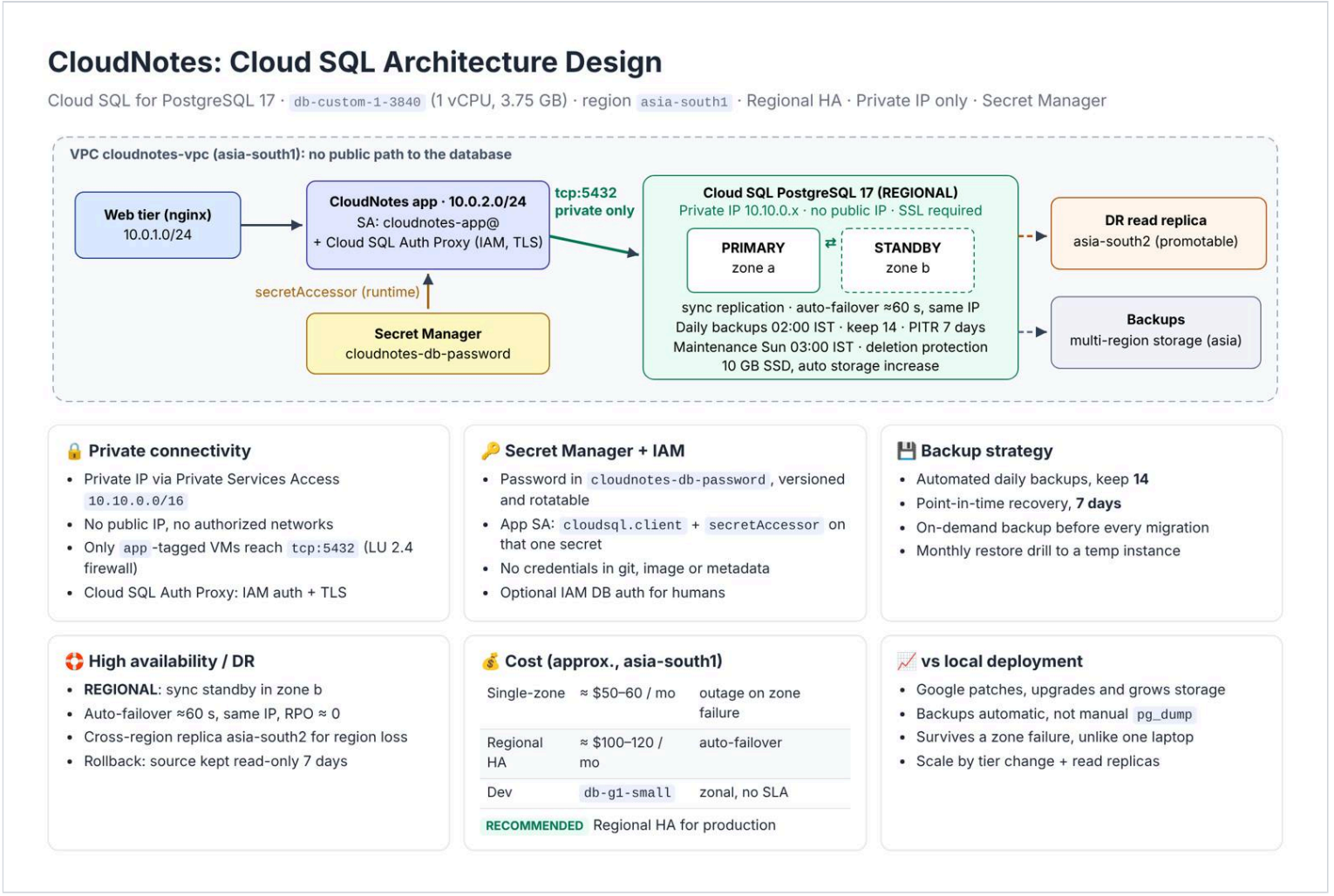
Caption: CloudNotes connects to PostgreSQL only through the DATABASE_URL environment variable. The real value lives in the git-ignored .env file, docker-compose.yml just passes the variable through, and the code reads it with os.environ.get. Repointing to the migrated database meant changing only the host to db-target. Passwords are masked, no credentials are hardcoded, and the database is reachable only on the private Docker network.

Screenshot 4 — CloudNotes Running on PostgreSQL



Caption: The app starts and its health check reports db_host=db-target, PostgreSQL 17. The migrated notes are returned by the app and shown in the browser, and a new note created after the migration was saved in the target database (row count went up by one) while the source stayed unchanged. This proves the application works end to end on the migrated database.

Screenshot 5 — Cloud SQL Architecture Design



Caption: In production the database moves to Cloud SQL for PostgreSQL 17. Reliability improves with automated daily backups (14 kept), 7-day point-in-time recovery, a Sunday maintenance window and a regional HA standby with automatic failover. Security improves with a private IP only, the Auth Proxy, and the password in Secret Manager read by an IAM service account. Scalability improves because Google handles patching and storage growth and the tier can grow. Regional HA (about 2x the cost of single-zone) is recommended for production.